NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-94/3K
6 August 1987
SUPERSEDING
MIL-R-94/3J
26 October 1981

PERFORMANCE SPECIFICATION

RESISTORS, VARIABLE, COMPOSITION, STYLE RV6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RV6 composition, variable, resistors.

2. APPLICABLE DOCUMENT

2.1 Government document.

2.1.1 Specification. Unless otherwise specified, the following specification of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-94 - Resistors, Variable, Composition, General Specification for.

(Copies of specification required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-94, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.2.1 Mounting and locking nuts. The mounting nut shall be .078-inch thick and shall measure .313-inch across the hexagonal flats. For locking bushings, the locking nut shall be .157-inch thick and shall measure .313 inch across the hexagonal flats. Thread size shall be .250-32NEF-2B.

3.2.2 Internal-tooth lockwasher. Internal-tooth lockwasher of suitable size shall be supplied and when mounted, shall have a maximum thickness of approximately 0.045 inch.

3.2.3 Retainer rings. If retainer rings are used, they shall be not thicker than 0.032 inch.

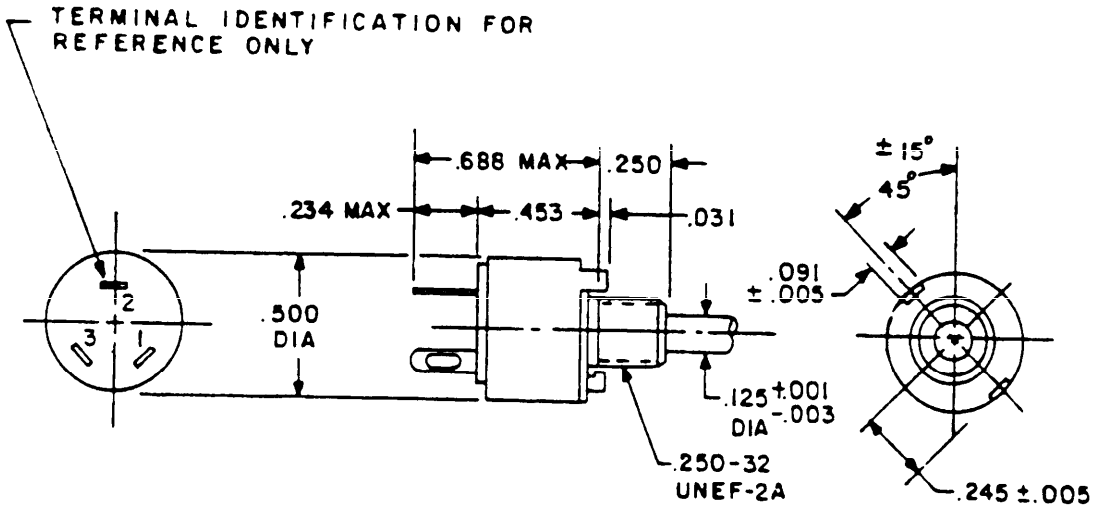
3.3 Total mechanical rotation. The total mechanical rotation without switch shall be within the limits of 292° and 298°.

3.4 Shaft length. The length of operating shall be in accordance with table I.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:55Z
Check the source to verify that this is the current version before use.

STANDARD-BUSHING TYPE (N)



LOCKING-BUSHING TYPE (L)

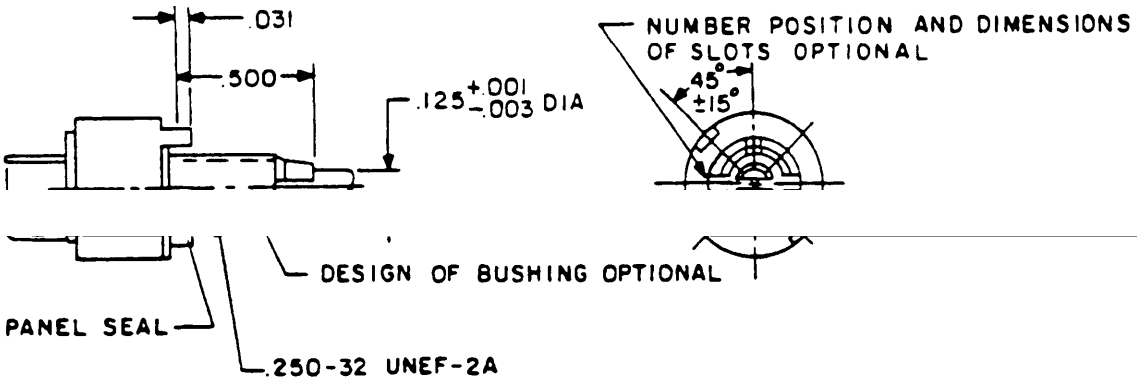
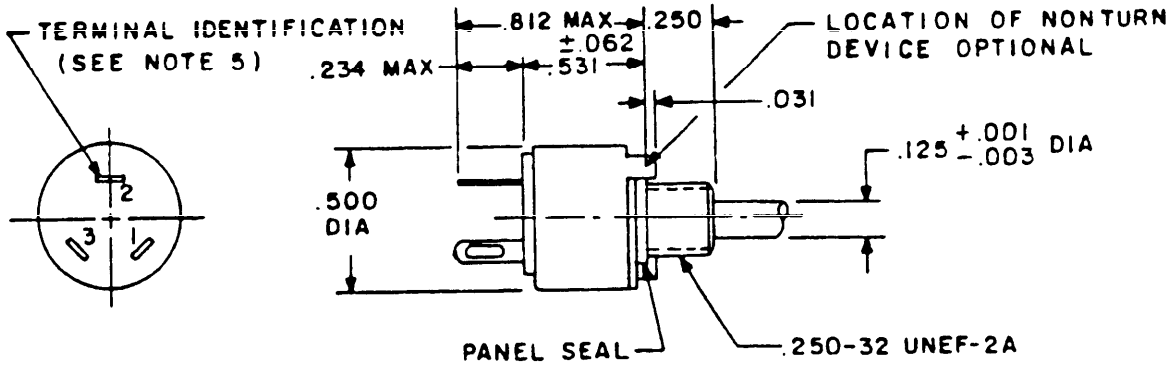
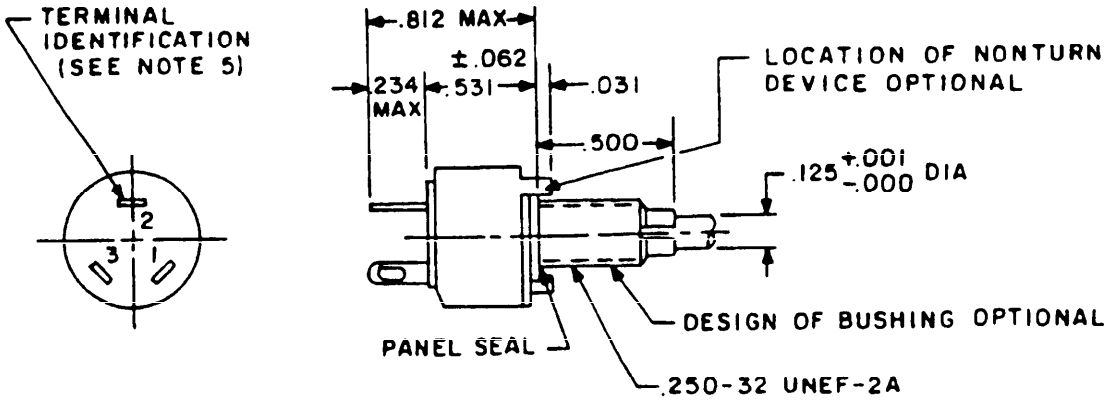


*FIGURE 1. Style RV6 resistors.

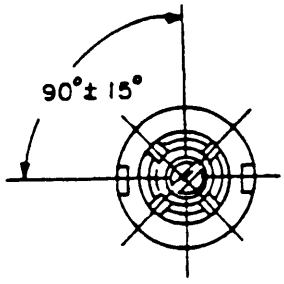
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:55Z
Check the source to verify that this is the current version before use.



SHAFT- AND PANEL-SEALED-STANDARD
BUSHING TYPE (S) (SEE NOTE 5)



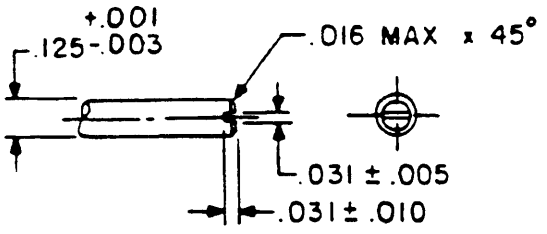
SHAFT AND PANEL-SEALED-LOCKING
BUSHING TYPE (T)



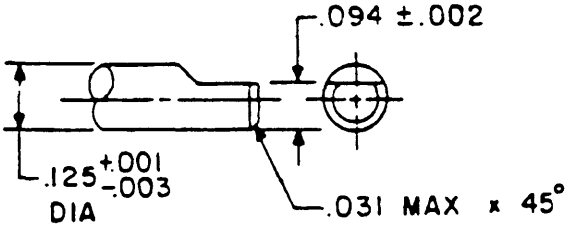
OPTIONAL LOCATION OF
NONTURN DEVICES

* FIGURE 1. Style RV6 resistors - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-16T04:55Z
Check the source to verify that this is the current version before use.



SLOTTED SHAFT



FLATTED SHAFT
(SEE NOTE 5)

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.062	1.57	.453	11.51
.002	0.05	.091	2.31	.500	12.70
.003	0.08	.094	2.39	.531	13.49
.005	0.13	.125	3.18	.688	17.43
.010	0.25	.234	5.94	.812	20.62
.016	0.41	.245	6.22		
.031	0.79	.250	6.35		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 (0.41 mm).
4. When terminals are located symmetrically, the contact terminal shall be identified on the unit. The identifying mark shall be at the option of the supplier.
5. Terminals shall be numbered when using the optional location of the nonturn device. When using the shaft and panel sealed standard bushing with the optional location of the nonturn device, the flatted shaft shall not be used.

* FIGURE 1. Style RV6 resistors - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261O-16T04:55Z
Check the source to verify that this is the current version before use.

TABLE I. Length of operating shaft. 1/

Symbol	Length of operating shaft from mounting surface of resistor (nominal)		
	Flatted	Slotted	
	Bushings N and S (inches, ±.313)	Bushings N and S (inches, ±.313)	Bushings L and T (inches, ±.313)
A	---	.625 2/	.625
B	---	.500 3/	---
D	.875	.875	.875 2/
E	---	---	---
G	---	---	---
L	---	.375	---

1/ For resistors with panel seals, the mounting surface shall be considered the metal face of the panel seal when the seal is firmly seated against the resistor body.
2/ For replacement purposes only. Not for new design.
3/ For Army and Air Force use only.

3.5 Nominal total resistance and rated continuous working voltage. The nominal total resistance and rated continuous working voltage shall be as specified in table II.

TABLE II. Nominal total resistance and rated continuous working voltage.

Nominal total resistance	RCWV 1/ (volts)		Nominal total resistance	RCWV 1/ (volts)	
	Taper A	Tapers C and F		Taper A	Tapers C and F
Ohms			Megohms		
100	7	5	.10	224	160
200 2/	10	7	.20 2/	316	200
250	11	8	.25	350	200
500	16	11	.50	350	200
1,000	22	16	1.0	350	200
2,000	31	22	2.0	350	200
2,500	35	25	2.5	350	200
5,000	50	36	5.0	350	200
10,000	71	50			
20,000 2/	100	70			
25,000	112	80			
50,000	158	112			

1/ Rated continuous working voltage at 70°C.
2/ For replacement purposes only. Not for new design.

3.6 Minimum resistance. The maximum value of minimum resistance shall be 15 ohms for total resistance values of 100 to 500 ohms inclusive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:55Z
Check the source to verify that this is the current version before use.

3.7 Power rating. The power rating shall be .500 watt for taper-A resistors and .250 watt for tapers-C and -F resistors.

3.8 Torque.

3.8.1 Operating. The torque required to effect rotation shall be not less than 0.50 ounce-inch nor more than 6 ounce-inches.

3.8.2 Stop. The torque applied through the operating shaft to the stops shall be 3 inch-pounds.

3.8.3 Locking bushing. For the locking-bushing type, the locking nut shall be tightened with a torque of 8 inch-pounds. After the locking nut is tightened, the contact arm shall not move when a torque of 20 ounce-inches is applied to the shaft.

3.9 Thermal cycling. The resistors shall not change in resistance in excess of 4 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-94, and as specified herein.

4.2 Dielectric withstanding voltage. In the dielectric withstanding voltage test, the applied potential shall be 750 volts rms at atmospheric pressure, and 350 volts rms at reduced barometric pressure.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-94.

6. NOTES

6.1 Notes. The notes specified in MIL-R-94 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1096)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:55Z
Check the source to verify that this is the current version before use.